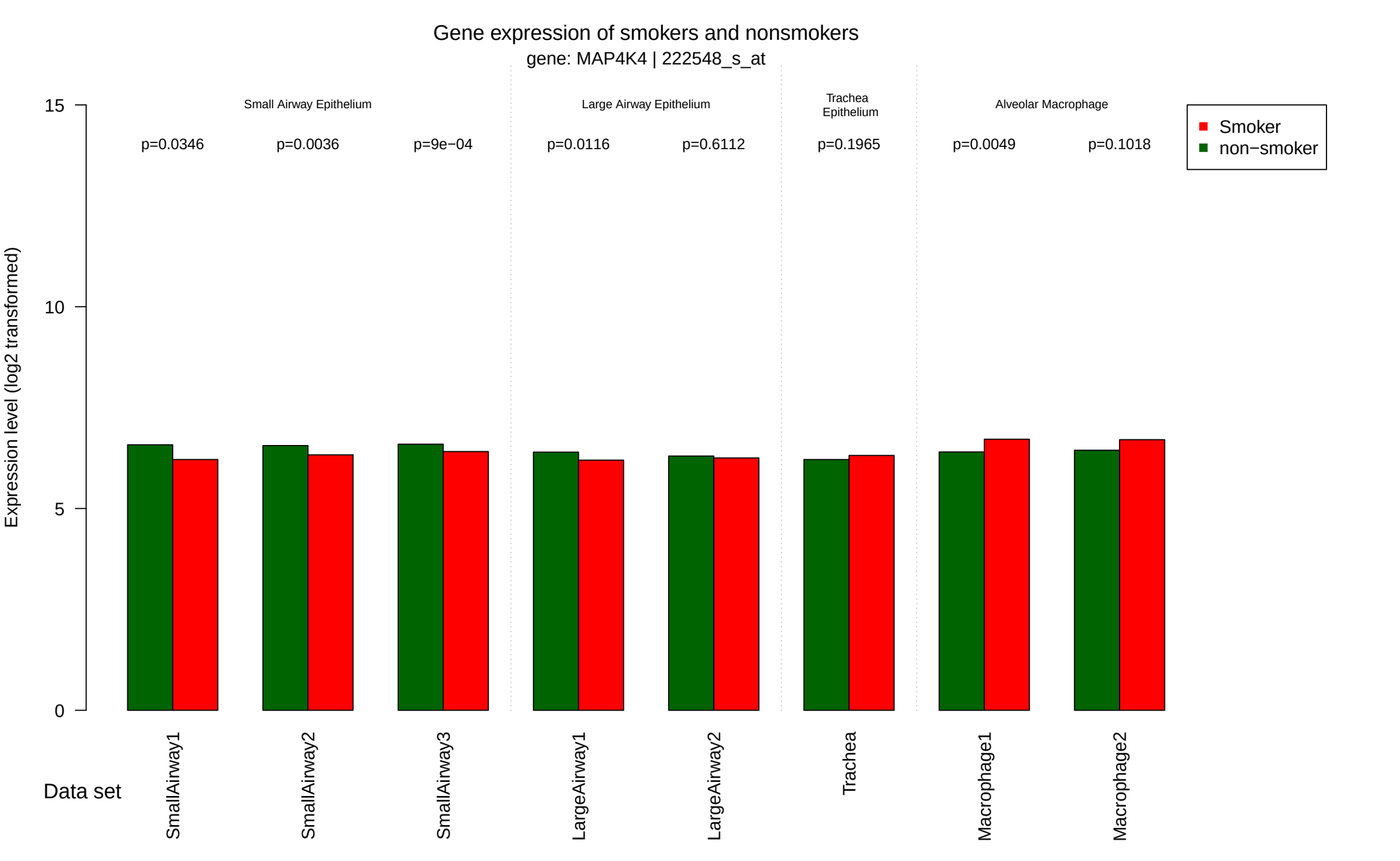
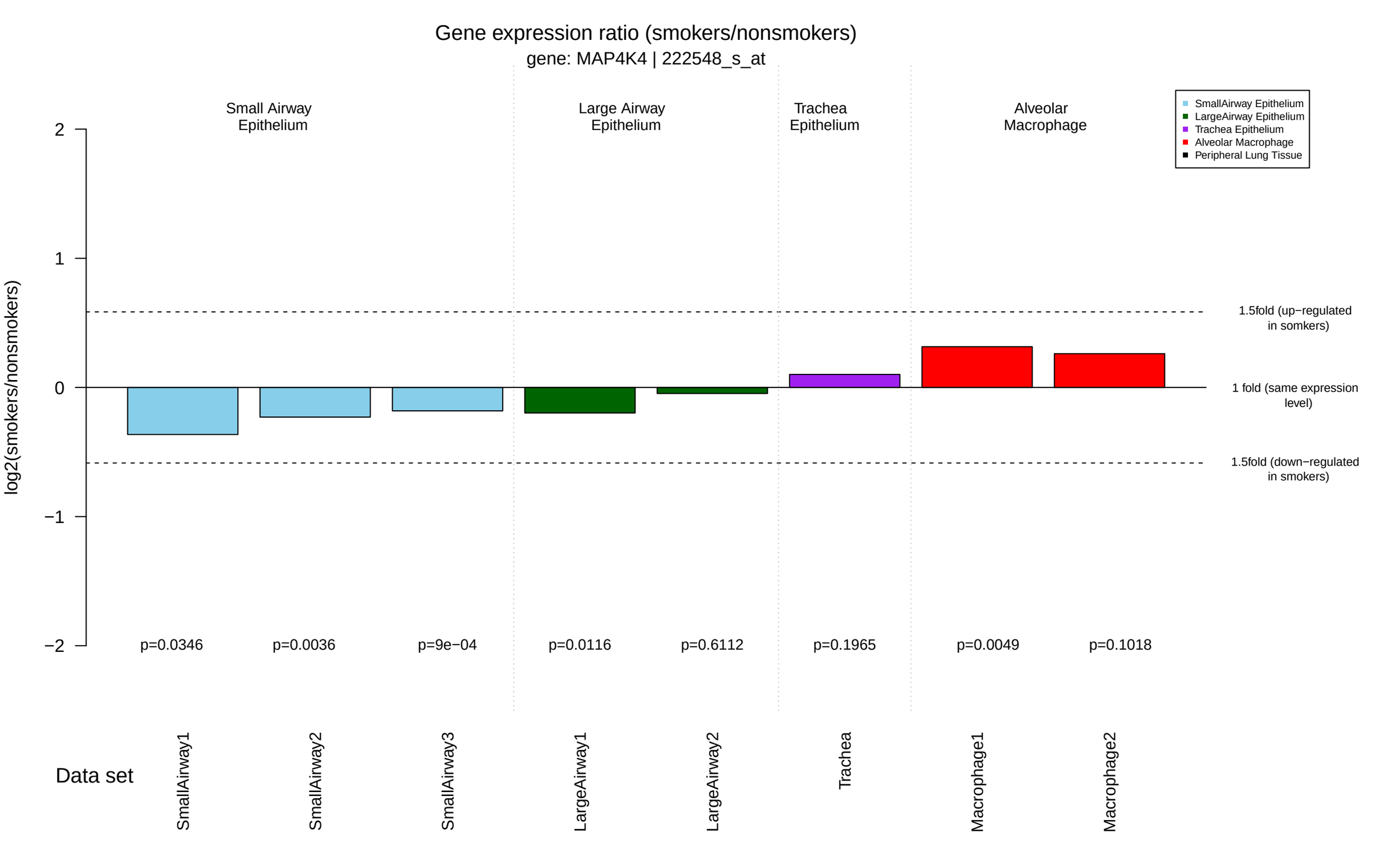
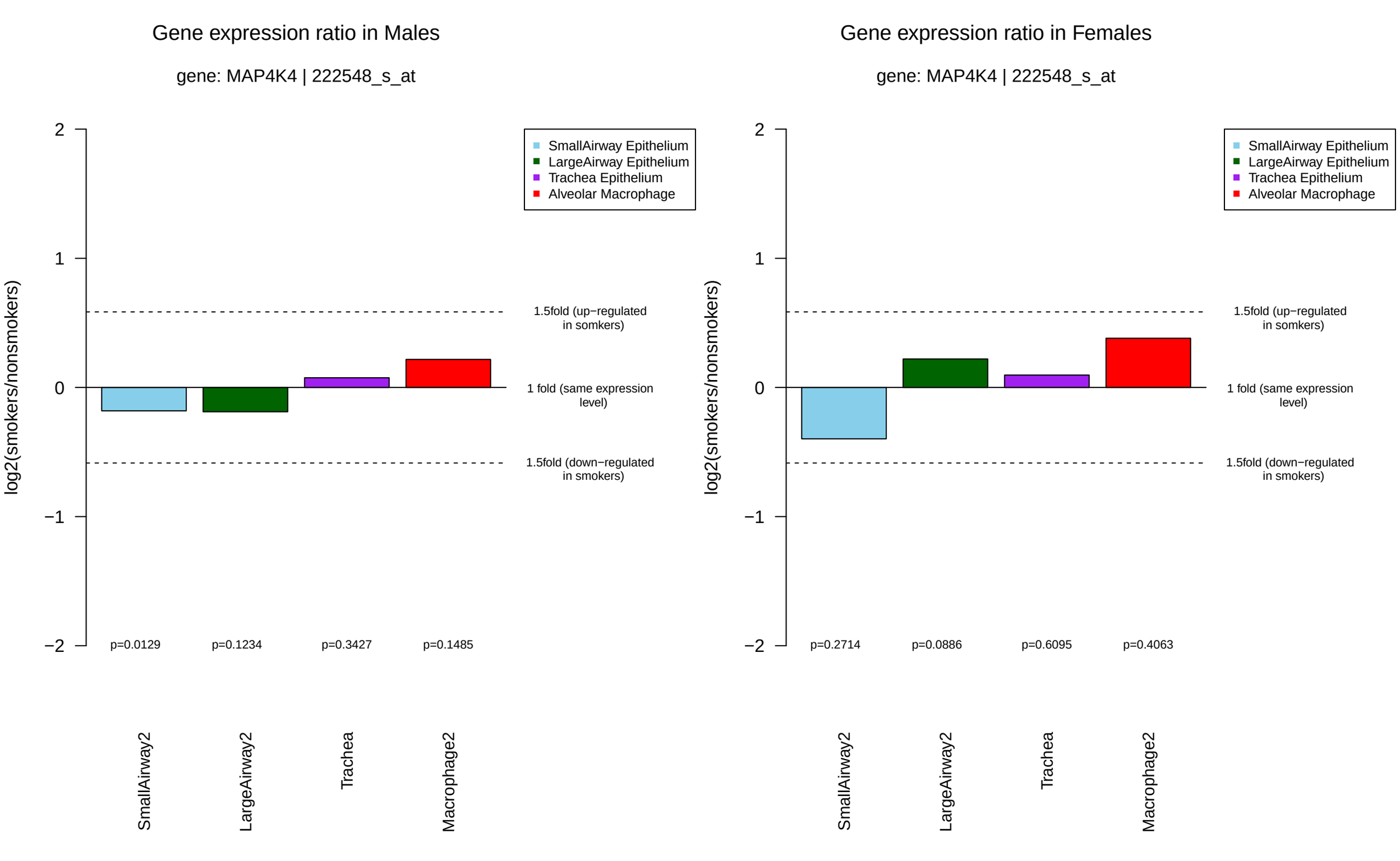
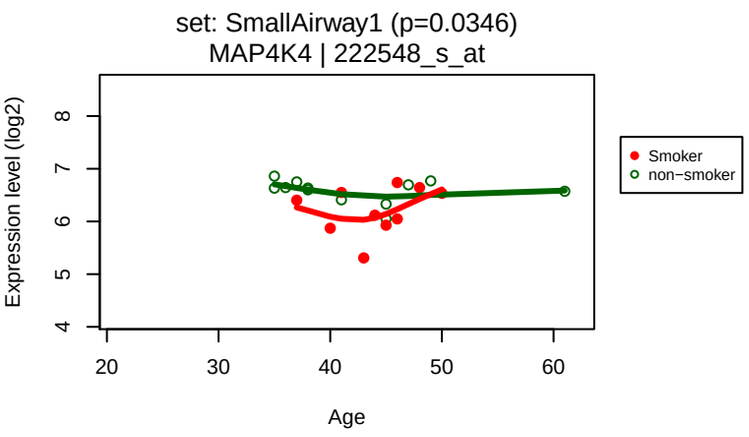
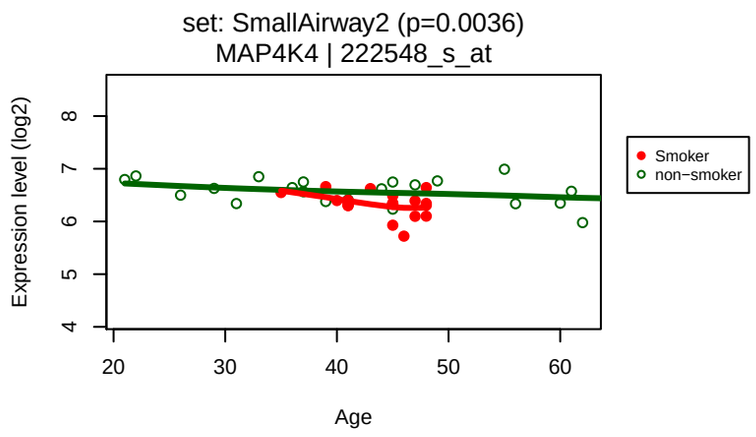
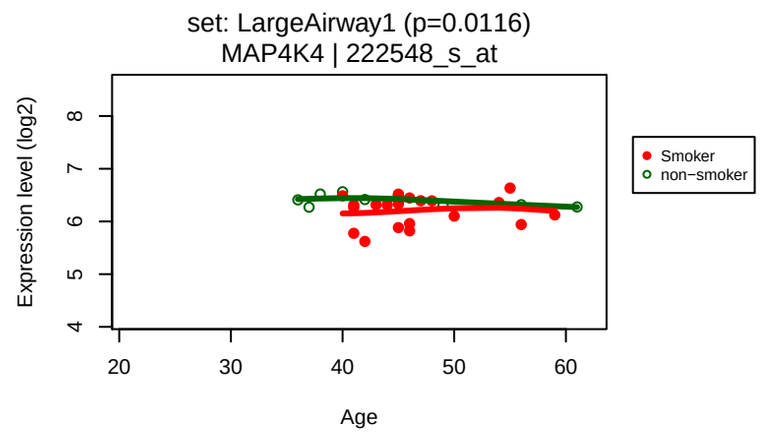

Gene expression of smokers and nonsmokers
gene: MAP4K4 | 222548_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0346
p=0.0036
p=9e−04
p=0.0116
p=0.6112
p=0.1965
p=0.0049
p=0.1018
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

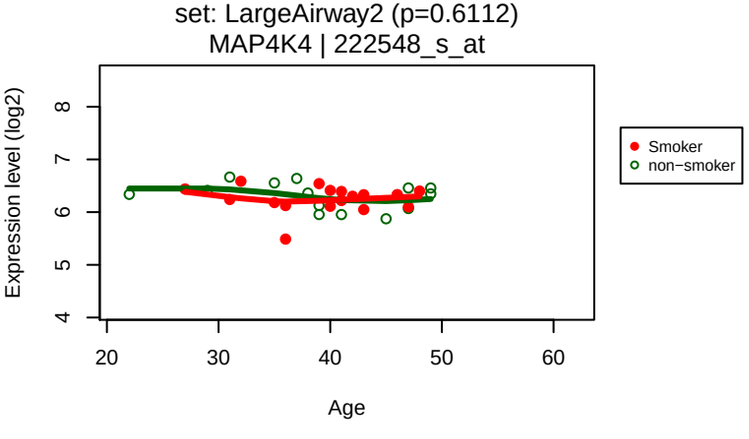

Gene expression ratio (smokers/nonsmokers)
gene: MAP4K4 | 222548_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0346
p=0.0036
p=9e−04
p=0.0116
p=0.6112
p=0.1965
p=0.0049
p=0.1018
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

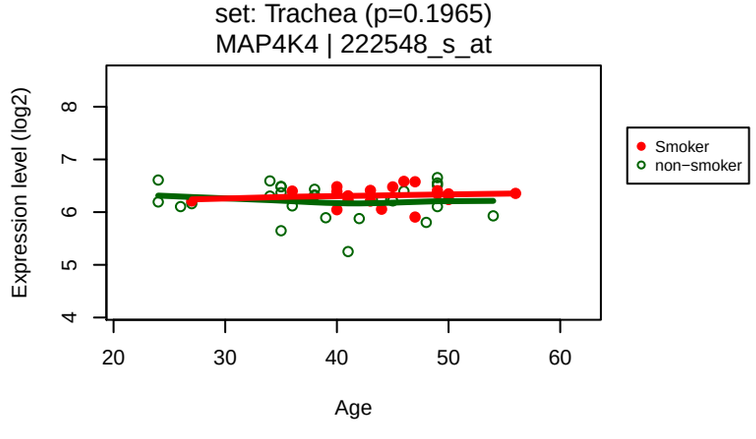

Gene expression ratio in Males
gene: MAP4K4 | 222548_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0129
p=0.1234
p=0.3427
p=0.1485
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP4K4 | 222548_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2714
p=0.0886
p=0.6095
p=0.4063
SmallAirway2
LargeAirway2
Trachea
Macrophage2

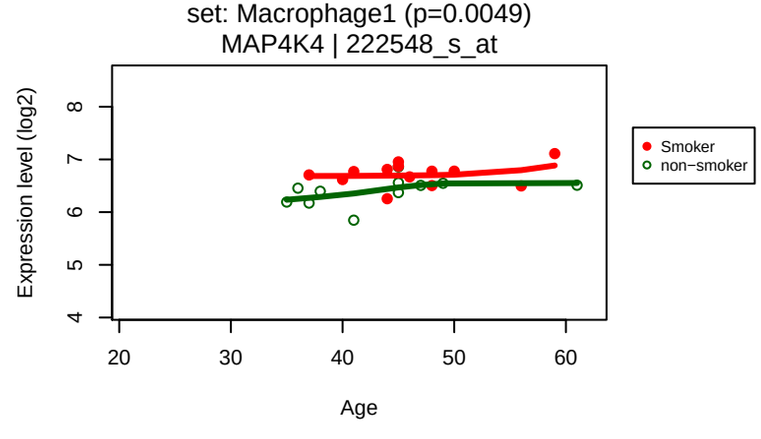

set: SmallAirway1 (p=0.0346)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker

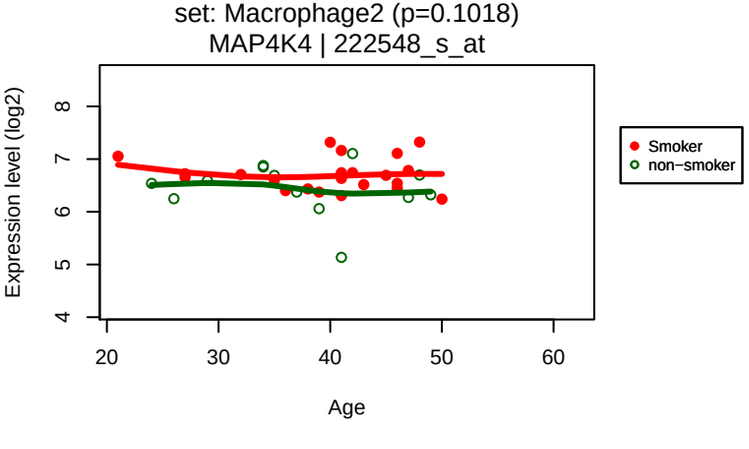

set: SmallAirway2 (p=0.0036)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker

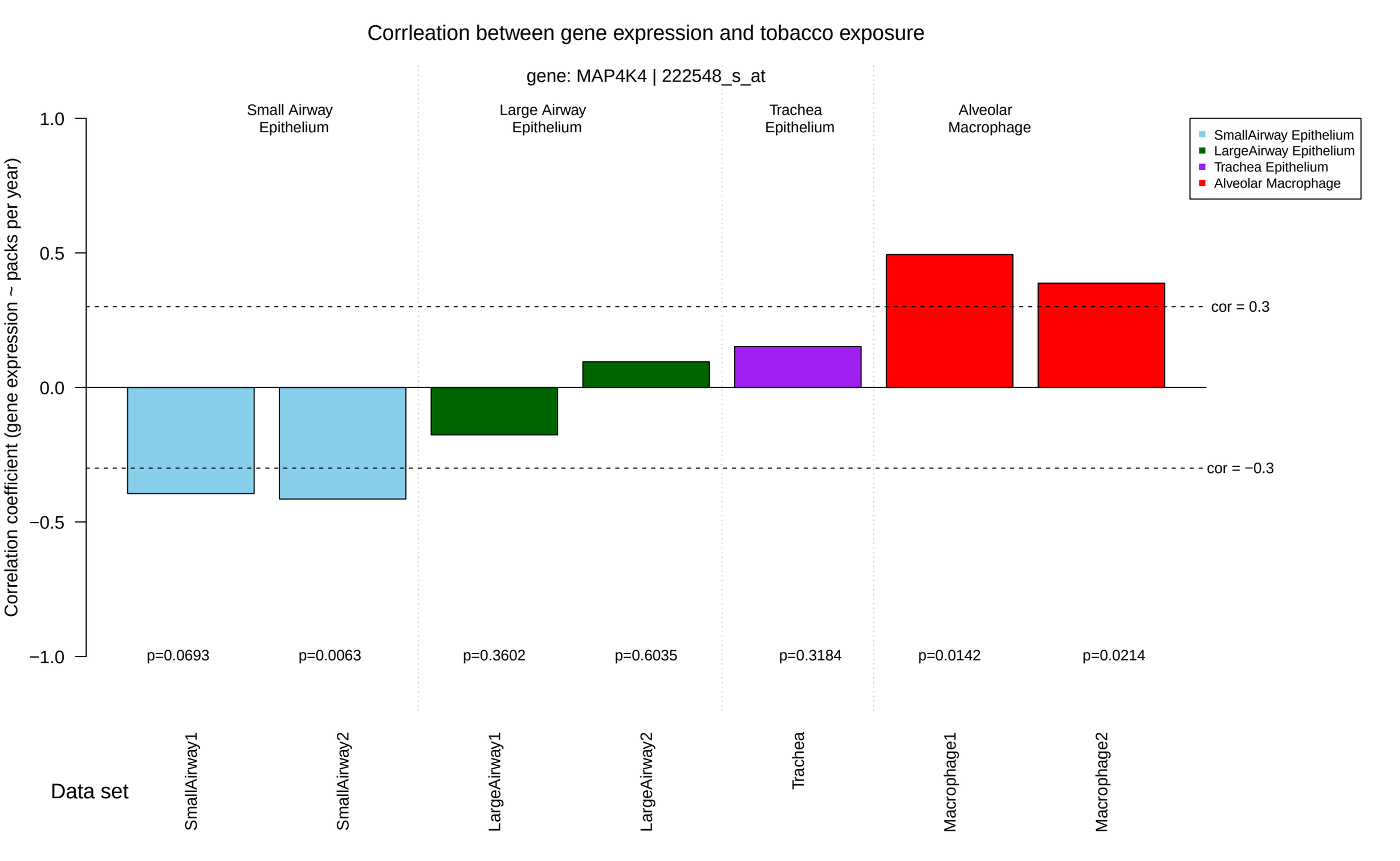

set: LargeAirway1 (p=0.0116)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6112)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1965)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0049)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1018)
MAP4K4 | 222548_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP4K4 | 222548_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0693
p=0.0063
p=0.3602
p=0.6035
p=0.3184
p=0.0142
p=0.0214
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2